

Chapter 3

Long-run Real Bond Parity

3.1 Introduction

Exchange rate determination remains an active puzzle (Meese and Rogoff (1983)). The empirical literature has sought to identify significant correlates to changes in exchange rates (Ferraro, Rossi, and Rogoff (2011), Verdelhan (2018), Jiang, Krishnamurthy, Lustig (2018)). The risk-based view of exchange rate determination has helped explain the cross section of currency portfolio returns and individual exchange rate variation (Lustig, Verdelhan (2009), Verdelhan (2018)), generating explainability of monthly, bilateral exchange rate movements as high as 80%.

This paper adopts the risk-based view of exchange rates. Following in the footsteps of Lustig et al. (2019), I make four assumptions: complete markets, the satisfaction of regularity conditions for a decomposition of the stochastic discount factor into a “permanent”, martingale component and a “transitory” component related to the inverse of long run real bond prices, a common “permanent”, martingale component across currencies, and the validity of the log of ten-year, real bond, monthly holding period returns as an approximation for the log of infinite-maturity, real bond, monthly holding period returns, to derive testable implications for the sign, and magnitude of the relationship between the log of real exchange rate changes and the log of long-maturity (i.e. ten-year) real bond returns. In particular, a positive shock to the log holding period return on the long-run foreign (domestic) real bond should be associated with contemporaneous, and one-to-one depreciation (appreciation) in the log of the foreign real exchange rate; once converted into a common numeraire, log holding period returns on real long-run bonds should be equal across currency baskets. In the style of Lustig et al. (2019) who derived this relation in nominal terms, I call this the “long, real bond return parity.”

The joint hypothesis is rejected in the post-2009 sample of developing and developed market currencies; however, I document a strong relationship between changes in the log real exchange rate and real foreign, and domestic holding period bond returns in the direction of parity.

3.2 Background

Definition 1. Let Assumptions I-IV refer to the following set of assumptions:

- I. Complete markets;
- II. Regularity conditions satisfied to decompose the stochastic discount factor in a permanent, martingale component and a transitory component related to the inverse of long run real bond prices;
- III. Common "permanent" martingale component across currencies;
- IV. Log of ten-year, real bond, monthly holding period returns approximate the log of infinite-maturity, real bond, monthly holding period returns.

Proposition 1. *When the Assumptions I-IV hold,*

$$\Delta q_{t+1} = hpr_{t+1}^{*,(10)} - hpr_{t+1}^{(10)} \quad (3.1)$$

where Δq_{t+1} is the log change in the real exchange rate between the domestic and foreign currency, $hpr_{t+1}^{*,(10)}$ is the log holding period return on the foreign, ten-year real bond, and $hpr_{t+1}^{(10)}$ is the log holding period return on the domestic, ten-year real bond.

Proof. Define a single currency pair as a domestic and foreign currency. To simplify the exposition, I will denote all domestic variables without stars, and foreign variables with stars. Suppose I have the stochastic discount factor M_{t+1} with the following representation in terms of the time $t + 1$ and t pricing kernels Λ_{t+1}, Λ_t

$$M_{t+1} = \frac{\Lambda_{t+1}}{\Lambda_t}. \quad (3.2)$$

Under regularity conditions, Alvarez and Jermann (2005) present a decomposition of the pricing kernel Λ_t into a martingale, permanent component Λ_t^P and a transitory component Λ_t^T using the price of a k maturity long-term bond $P_t^{(k)}$

$$\Lambda_t = \Lambda_t^P \Lambda_t^T, \text{ where } \Lambda_t^T = \lim_{k \rightarrow \infty} \frac{\delta^{t+k}}{P_t^{(k)}}. \quad (3.3)$$

The holding period return from time t to $t+1$ on the infinite maturity bond, which I denote by $HPR_{t+1}^{(\infty)}$, is the ratio of the transitory components at time t and $t+1$

$$HPR_{t+1}^{(\infty)} = \lim_{k \rightarrow \infty} HPR_{t+1}^{(k)} = \lim_{k \rightarrow \infty} \frac{P_{t+1}^{(k-1)}}{P_t^{(k)}} = \frac{\Lambda_t^T}{\Lambda_{t+1}^T}. \quad (3.4)$$

The corresponding log holding period return, which I denote by $hpr_{t+1}^{i,(\infty)}$ is

$$hpr_{t+1}^{i,(\infty)} = -\lambda_{t+1}^{i,T} + \lambda_t^{i,T}. \quad (3.5)$$

If markets are complete, there exists a unique, real stochastic discount factor. This factor generates a set of unique, currency-specific factors that differ in the real numeraire that they reference, M_{t+1}, M_{t+1}^* . The Alvarez and Jermann representation of M_{t+1} gives us

$$M_{t+1} = \frac{\Lambda_{t+1}}{\Lambda_t} = \frac{\Lambda_{t+1}^P \Lambda_{t+1}^T}{\Lambda_t^P \Lambda_t^T}, \text{ where } \Lambda_t^T = \lim_{k \rightarrow \infty} \frac{\delta^{t+k}}{P_t^{(k)}}. \quad (3.6)$$

Let Q_{t+1} denote the time $t+1$ real exchange rate, defined as units of the foreign real goods basket per unit of the domestic real goods basket

$$Q_{t+1} = \frac{\text{Units of foreign real goods baskets}_{t+1}}{\text{Domestic real goods basket}}. \quad (3.7)$$

An accounting identity gives us

$$M_{t+1} = M_{t+1}^* \frac{Q_{t+1}}{Q_t}. \quad (3.8)$$

This is from the Euler equations of the stand-in domestic and foreign investors: for an asset return R^* expressed in foreign currency basket terms $\mathbb{E}_t[M_{t+1} R_{t+1}^* \frac{Q_t}{Q_{t+1}}] = 1$ and $\mathbb{E}_t[M_{t+1}^* R_{t+1}^*] = 1$. State-by-state equality is an accounting identity that holds under no arbitrage. In logs, this corresponds to

$$\Delta q_{t+1} = m_{t+1} - m_{t+1}^*. \quad (3.9)$$

The AJ decomposition gives us

$$\Delta q_{t+1} = [(\lambda_{t+1}^T - \lambda_t^T) + (\lambda_{t+1}^P - \lambda_t^P)] - [(\lambda_{t+1}^{T*} - \lambda_t^{T*}) + (\lambda_{t+1}^{P*} - \lambda_t^{P*})]. \quad (3.10)$$

Lustig et al. (2019) show that average excess returns on the carry trade using long-run bonds and with a three-month horizon, is zero, suggesting that the entropies of the permanent

components are equal.¹ If currency SDFs share a common permanent (martingale) $\lambda_t^{P*} = \lambda_t^P$, this restriction is satisfied and

$$\Delta q_{t+1} = -(\lambda_{t+1}^{T*} - \lambda_t^{T*}) + (\lambda_{t+1}^T - \lambda_t^T) \quad (3.11)$$

$$= hpr_{t+1}^{*,(\infty)} - hpr_{t+1}^{(\infty)}. \quad (3.12)$$

□

This is a statement about realizations not expectations. This is not a novel derivation. Lustig et al. (2019) are the first to derive it in nominal terms in their appendix. This paper rewrote it in real terms, and will test it.

Let t index time and i the foreign currency, this directly motivates a regression of the form

$$\Delta q_{t+1} = \alpha + \beta_1 hpr_{t+1}^{*,(\infty)} + \beta_2 hpr_{t+1}^{(\infty)} + \epsilon_t \quad (3.13)$$

where Δq_{t+1} is the log change in the real exchange rate, $hpr_{t+1}^{*,(\infty)}$ is the log holding period return on the foreign, infinite maturity real bond, and $hpr_{t+1}^{(\infty)}$ is the log holding period return on the domestic, infinite maturity real bond.

However, I face a practical problem. There are no infinite maturity real, risk-free bonds. Therefore, I assume that the log of ten-year bond returns approximate the log of infinite bond returns; that is, that

$$\Delta q_{t+1} = \alpha + \beta_1 hpr_{t+1}^{*,(10)} + \beta_2 hpr_{t+1}^{(10)} + \epsilon_t \quad (3.14)$$

is closely approximated by

$$\Delta q_{t+1} = \alpha + \beta_1 hpr_{t+1}^{*,(10)} + \beta_2 hpr_{t+1}^{(10)} + \epsilon_t \quad (3.15)$$

where $hpr_{t+1}^{*,(10)}$ is the log holding period return on the foreign, ten-year real bond, and $hpr_{t+1}^{(10)}$ is the log holding period return on the domestic, ten-year real bond.

3.3 Data

In this section, I describe the notation and the data.

¹Their result is in nominal terms and on currency portfolios. They argue that the inflation characteristics of portfolios suggest the result holds also in real terms. I find suggestive evidence that this is true when I look at the individual currencies in the sample. Interested readers may find the analysis in the Appendix.

3.3.1 Notation

This section describes notation for exchange rates, bond prices, and currency and bond returns.

Currency and national price levels

Let S_{t+1} denote the time $t + 1$ nominal exchange rate, defined as units of foreign currency per unit of the domestic currency

$$S_{t+1} = \frac{\text{Units of foreign currency}_{t+1}}{\text{Unit of domestic currency}}. \quad (3.16)$$

Let Z_{t+1} denote the time $t + 1$ domestic national price level, defined as the domestic nominal price of one unit of the domestic real goods basket

$$Z_{t+1} = \frac{\text{Units of domestic currency}_{t+1}}{\text{Domestic real goods basket}}. \quad (3.17)$$

Let Π_{t+1} denote gross inflation from time t to time $t + 1$ in the domestic currency

$$\Pi_{t+1} = \frac{Z_{t+1}}{Z_t}. \quad (3.18)$$

National price levels are vis-a-vis currency specific real goods baskets; this is to reflect cross-country differences in what governments consider a basket of real goods when reporting national price levels.

I now define other objects defined vis-a-vis these variables.

Let Q_{t+1} denote the time $t + 1$ real exchange rate, defined as units of the foreign real goods basket per unit of the domestic real goods basket

$$Q_{t+1} = \frac{\text{Units of foreign real goods baskets}_{t+1}}{\text{Domestic real goods basket}} = S_{t+1} \frac{Z_{t+1}}{Z_{t+1}^*}. \quad (3.19)$$

Let Π_{t+1} denote gross inflation from time t to time $t + 1$ in the domestic currency

$$\Pi_{t+1} = \frac{Z_{t+1}}{Z_t}, \quad (3.20)$$

and Π_{t+1}^* denote gross inflation from time t to time $t + 1$ in the foreign currency

$$\Pi_{t+1}^* = \frac{Z_{t+1}^*}{Z_t^*}. \quad (3.21)$$

Following the convention of denoting by the lowercase the logs of variables, let Δq_{t+1} denote the log change in real exchange rates from time t to time $t + 1$

$$\Delta q_{t+1} = \Delta s_{t+1} + \pi_{t+1} - \pi_{t+1}^*. \quad (3.22)$$

Bonds

Let $P_t^{(N)}$ denote the zero-coupon, N -maturity, risk-free real bond which pays a unit of the real basket at maturity.

Let $HPR_{t+1}^{(N)}$ denote the gross, holding period return on the N -maturity zero coupon real bond from time t to time $t + 1$

$$HPR_{t+1}^{(N)} = \frac{P_{t+1}^{(N-1)}}{P_t^{(N)}} \quad (3.23)$$

and $hpr_{t+1}^{(N)}$ denote its log counterpart

$$hpr_{t+1}^{(N)} = \log(HPR_{t+1}^{(N)}). \quad (3.24)$$

For example, the log, monthly holding period return on a ten-year bond, which I denote by $hpr_{t+1}^{(10)}$, is

$$hpr_{t+1}^{(10)} = \log(HPR_{t+1}^{(10)}). \quad (3.25)$$

$$= \log\left(\frac{P_{t+1}^{(9\frac{11}{12})}}{P_t^{(10)}}\right). \quad (3.26)$$

To compute holding period returns, I need zero-coupon real bonds prices at the 10, $9\frac{11}{12}$ -year maturities.

Let M_{t+N} be the nominal stochastic discount factor that prices $t + N$ cash flows in domestic currency. Consider a zero-coupon, N -maturity risk-free nominal bond which pays a unit of currency at maturity. Its price is $B_t^{(N)}$ and its zero rate is $y_t^{(N)}$

$$B_t^{(N)} = E_t[M_{t+N}] \quad (3.27)$$

$$\equiv \frac{1}{(1 + y_t^{(N)})^N}. \quad (3.28)$$

Consider a zero-coupon, N -maturity inflation swap. At maturity, the fixed leg investor will pay the fixed rate $i_t^{(N)}$ on the notional; the floating leg investor will pay the realized change in the referenced inflation index series $\frac{Z_{t+N}}{Z_t}$. Inflation swaps are priced at par: the fixed leg is such that the fixed and floating legs have the same value, and that the contract value is zero. This is expressed as

$$0 = E_t[M_{t+N} \times (\frac{Z_{t+N}}{Z_t} - (1 + i_t^{(N)})^N)]. \quad (3.29)$$

Consider a zero-coupon, N -maturity, risk-free real bond which pays a unit of the real basket at maturity. Its price is $P_t^{(N)}$.

Using the previous expression, I can express $P_t^{(N)}$, which is often not directly observable, as a function of $y_t^{(N)}$, $i_t^{(N)}$, which I often do observe directly:

$$0 = E_t[M_{t+N} \times (\frac{Z_{t+N}}{Z_t} - (1 + i_t^{(N)})^N)] \quad (3.30)$$

$$E_t[M_{t+N} \frac{Z_{t+N}}{Z_t}] = E_t[M_{t+N} \times (1 + i_t^{(N)})^N] \quad (3.31)$$

$$E_t[M_{t+N} \frac{Z_{t+N}}{Z_t}] = E_t[M_{t+N}](1 + i_t^{(N)})^N \quad (3.32)$$

$$P_t^{(N)} = E_t[M_{t+N}](1 + i_t^{(N)})^N, \text{ since } P_t^{(N)} \equiv E_t[M_{t+N} \frac{Z_{t+N}}{Z_t}] \quad (3.33)$$

$$P_t^{(N)} = \left[\frac{(1 + i_t^{(N)})}{(1 + y_t^{(N)})} \right]^N \quad (3.34)$$

3.3.2 Data Sources

I turn to the paper's data. I first describe the dataset, and then discuss the data selection process and features of the data relevant to the interpretation of the results.

Data sources

The data is a panel of monthly national price levels and daily closing prices for zero-coupon nominal overnight index swap (OIS) and zero-coupon inflation swaps, nominal and real bonds, nominal exchange rates, and nominal forward exchange rates for eight currencies. Six of the eight are from developed market economies: the Japanese yen, euro, Swedish krona, British pound, U.S. dollar, Australian dollar; the remaining two are from developing market economies: South African rand, and Brazilian real²

²I listed the currencies in order of the average level of their three-month, nominal interest rates. I report results in this order as well. This was motivated by the fact that the failure of UIP is a key stylized fact, and so currencies are often thought of in terms of their nominal short interest rate levels.

All data are obtained from Bloomberg at the daily frequency for the period from January 01, 2009 to December 31, 2020. Data are aggregated to the monthly frequency, which corresponds to the most common release frequency of national price level series.

My dataset is constrained by data availability as few currencies have developed markets for long-maturity inflation swaps, real bonds, and overnight index swaps³

Zero-coupon inflation and OIS rates, and zero-coupon nominal and government bond rates were obtained at the three-month, and one, two, five, nine and ten year maturities. With the exception of inflation swap rates, rates are rates derived by Bloomberg from traded prices.

In an OIS swap, a fixed rate of interest decided at contract birth is exchanged for a floating rate of interest over the swap’s lifetime. The swap rate quote is the rate on the fixed leg of the swap.⁴ Table 3.3.2 lists the series obtained, and their source.

Table 3.1: Unbalanced sample data sources and coverage

Data series	Source
Zero-coupon inflation swap rates	Bloomberg
Zero-coupon OIS rates	Bloomberg
Zero-coupon nominal and inflation-indexed government bond rates	Bloomberg
Nominal exchange rates	Bloomberg
Nominal forward exchange rates	Bloomberg, author’s calculations
National price levels	Bloomberg

Notes: This table describes the sources and coverage of the data. The data covers a small, heterogeneous panel of developed and developing market currencies: Japanese yen, euro, Swedish krona, British pound, U.S. dollar, Australian dollar, South African rand, and Brazilian real. All data are obtained at the daily frequency for the period from January 01, 2009 to December 31, 2020. Data were aggregated to the monthly frequency as this corresponds to the most common release frequency of national price level series. Nominal forward exchange rates for maturities longer than one year was constructed as in Du et al. (2018).

In Table 3.3.2 I report summary statistics for the main data series by currency. Panel A reports the mean, standard error, and standard deviation of the ten-year nominal rate $y_t^{*,(10)}$, the ten-year real rate $r_t^{*,(10)}$, the annual year-over-year inflation rate π_t , and the ten-year

³While inflation indexed bonds have existed before 2009, limited issuance across the maturity structure and embedded options make it difficult to reliably bootstrap the real bond curve for the currency set. Long-maturity inflation swap data exist for 2004 onwards, but long-maturity OIS data do not exist until 2009. The choice of OIS as the benchmark is motivated by the TIPS-Treasury puzzle, which indicates “mispricing” in real and/or nominal government cash bond markets in the U.S. (Fleming et al. (2014)), Japan, U.K., Germany (Kita and Tortice, 2019).

⁴Pricing errors, if they mimic classical measurement error, should minimally affect the cross sectional empirics if the variance of the error is small and the sample large enough; for the time series empirics, they will bias coefficients downwards towards zero.

inflation compensation rate $i_t^{*,(10)}$. Panel B reports the same statistics for the one-year real rate $r_t^{*,(1)}$, and the slope of the real rate curve which is the difference between the ten-year and one-year real rates $r_t^{*,(10)} - r_t^{*,(1)}$.

The table paints a picture of a currency sample with appreciable heterogeneity along the recorded dimensions. The currencies with the lowest and highest average nominal rates at the ten-year maturity are, respectively, the Japanese yen at 0.62% and the Brazilian real at 10.4%, a difference of nearly ten percentage points. The range of average real rates is more narrow, but also appreciable. The currencies with lowest and highest real rates at the ten-year maturity is, respectively, the British pound at -1.12% and the Brazilian real 5.11%. The sharply negative real rate for the British pound, which is below even that for the Japanese yen, underscores the unprecedented nature of the sample's low nominal and real interest rate environment in the context of recorded history⁵. This warrants cautious interpretation and extrapolation of the empirical results.

On the inflation front, developed market currencies tend to have lower and less variable inflation than developing market currencies. Notably, however, the standard deviation of inflation for the British pound is 1.27%, which is similar in magnitude to the corresponding number for the South African rand. The average level of inflation swap rates generally increase with the level of inflation. The standard deviation on realized annual inflation is larger than the standard deviation on ten-year inflation swaps, which is suggestive of shocks in the inflation process that net out over a ten-year horizon. I emphasize that inflation swap rates are objects that are related to but not equal to either the inflation risk premia, or the holding period return on inflation swaps, the latter of which has a direct relation to the holding period returns on real bonds.

Turning to the yield curve, one-year real yields are negative for all developed market currencies, except the Australian dollar and there is no clear relationship between the level of one-year real yields and the slope of the real yield curve. One possible reason is that statistical relation between these two objects is more obscure when levels of real yields are so low and close across currencies. One-year real yields are positive for the two developing market currencies and are higher the steeper the slope of the real yield curve.

⁵The United Kingdom's decision to leave the European Union in June 2016 precipitated a sharp exchange rate 10% currency depreciation and a concomitant positive shock to inflation.

Table 3.2: Summary statistics for the cross section

(in percentage points)	Currency							
	British pound	Japanese yen	euro	Swedish krona	U.S. dollar	Australian dollar	South African rand	Brazilian real
	Panel A: Summary statistics							
$y_t^{*,(10)}$	Mean	2.13	0.62	1.52	1.88	2.41	3.59	8.17
	s.e.	[0.09]	[0.04]	[0.10]	[0.09]	[0.07]	[0.13]	[0.24]
	s.d.	1.09	0.48	1.19	1.08	0.81	1.61	2.46
$r_t^{*,(10)}$	Mean	-1.12	0.32	-0.09	-0.14	0.16	1.09	1.58
	s.e.	[0.09]	[0.07]	[0.07]	[0.09]	[0.05]	[0.10]	[0.12]
	s.d.	1.09	0.83	0.79	1.02	0.63	1.20	0.66
π_t	Mean	2.87	0.40	1.15	1.04	1.70	1.99	4.87
	s.e.	[0.11]	[0.09]	[0.08]	[0.09]	[0.08]	[0.07]	[0.44]
	s.d.	1.27	0.99	0.94	0.98	0.87	0.80	1.13
$i_t^{*,(10)}$	Mean	3.29	0.31	1.61	2.02	2.25	2.47	6.50
	s.e.	[0.02]	[0.05]	[0.04]	[0.02]	[0.03]	[0.04]	[0.12]
	s.d.	0.22	0.56	0.44	0.27	0.34	0.45	1.21
Panel B: Yield curve characteristics								
$r_t^{*,(1)}$	Mean	-1.93	-0.04	-0.70	-0.72	-0.48	0.42	0.85
	s.e.	[0.08]	[0.10]	[0.05]	[0.06]	[0.08]	[0.08]	[0.20]
	s.d.	0.91	1.14	0.64	0.67	0.92	0.96	1.03
$r_t^{*,(10)} - r_t^{*,(1)}$	Mean	0.81	0.36	0.61	0.58	0.64	0.67	0.72
	s.e.	[0.10]	[0.05]	[0.05]	[0.06]	[0.08]	[0.04]	[0.12]
	s.d.	1.21	0.61	0.63	0.76	0.91	0.51	0.63

Notes: This table displays the mean, standard error of the mean, the standard deviation of the ten-year nominal rate $y_t^{*,(10)}$, the ten-year real rate $r_t^{*,(10)}$, the annual inflation rate π_t , and the ten-year inflation compensation rate $i_t^{*,(10)}$, the one-year real rate based $r_t^{*,(1)}$, and the slope of the real rate curve which is the difference between the ten-year and one-year real rates $r_t^{*,(10)} - r_t^{*,(1)}$.

While I make the best use of available data, I pre-emptively emphasize that features of the dataset warrant cautious interpretation of the results. First, the dataset is limited in size. The dataset begin in January 2009, and the first return observation is for February 2009. Currencies have monthly observations that cover anywhere from eight to twelve years, which translates to 101 to 143 observations per currency, tenor pair. Observations are all in the post-crisis period during which I have seen historically unprecedented negative rates and central bank intervention in the long maturity bond market. This suggests the results should be cautiously interpreted within the context of the sample period they are based on. Furthermore, I have a small, and heterogeneous cross section, which restricts the analysis to individual currencies, as opposed to currency portfolios in which I might be able to appeal to diversification to isolate currency factors.

Data Sources

To construct the ten-year holding period returns on real bonds, I require zero-coupon real bond prices/rates. I constructed these using zero coupon, risk-free nominal OIS rates and zero-coupon inflation swap rates, except for Brazilian real for which I used the implied zero-coupon inflation swap rate from real bond prices.

By the law of one price, the price of the N -maturity, zero coupon domestic risk-free real bond $P_t^{(N)}$ can be expressed as a function of the corresponding zero-coupon inflation swap $i_t^{(N)}$ and nominal bond rate $y_t^{(N)}$

$$P_t^{(N)} = \frac{(1 + i_t^{(N)})^N}{(1 + y_t^{(N)})^N}. \quad (3.35)$$

One concern is that this relation may not strictly hold, and therefore, that the real bond price cannot be replicated in this way. In the United States, Fleckstein, Longstaff, and Lustig (2009) find that sovereign bonds are overpriced relative to their inflation indexed counterparts TIPS, by a weighted average of \$2.92, but with significant variation, and a maximum in their sample of nearly \$10. They provide suggestive evidence that this is not due to mispricing in the inflation swaps market: in the corporate bond market, nominal and inflation indexed bond mispricing is minimal, with an average of 1-7 basis points, with standard errors of 60-70 basis points. Even though they do not explicitly account for options, accounting for them would go in the opposite direction—nominal sovereign bonds would be even more overpriced. Kita and Tortorice (2019) document mispricing of similar size and the same direction in the sovereign bond markets for Japan and Germany. Whether the relative mispricing exists because of distortions in the price level of Treasuries, TIPS, or both from

their levels in the envisioned, background no arbitrage world is unclear.

Zero coupon nominal rates As the zero coupon nominal rates I use the discount rates implied by overnight index swaps (OIS). An OIS is a swap with floating payments based on a reference rate for unsecured overnight funding and paid at fixed intervals.⁶ Over the lifetime of a contract, counterparties post collateral to keep the net value of the swap contract at zero. In early 2000, the International Swaps and Derivatives Association (ISDA), a global trade organization of participants in the market for over the counter derivatives, conducted a survey that found more than 65% of plain vanilla derivatives, especially interest rate swaps, were collateralized according to the legal standard for collateralization set out in the Credit Support Annex (CSA), a legal document of recommendations in ISDA’s master agreement which regulates collateral for derivatives transactions; moreover, they were bilateral in that both counterparties, not just the lower rated counterparty posted collateral (Johannes and Sundaresan (2006)). Thus, the OIS is only exposed to overnight credit risk and is broadly considered “risk free.”⁷ Because the OIS contract is approximately, not completely, risk free, the rate itself is influenced by the credit risk of the refreshed panel of participants in unsecured overnight funding markets; however, this effect of credit risk should be limited by the fact that the risk is for the portion of the contract that may be undercollateralized because of overnight moves in the market.⁸ Hull and White (2015) argue that OIS is the best proxy for the risk-free rate; furthermore, post-crisis, practitioners have switched from using the LIBOR zero curve to the OIS zero curve for discounting. The use of OIS constrains the sample size as data on long maturity swaps are only available on Bloomberg after 2009.

In the developed markets, one may be concerned that the “risk-free approximation” error is non-trivial. To alleviate this concern, for the South African rand and the Brazilian real, I construct synthetic OIS rates equal to the OIS rates that would hold if the domestic OIS rate was “risk-free” and covered interest parity held. Let $y_t^{(N)}$ denote the time t , maturity N nominal, domestic OIS rate. Let $\rho_t^{(N)}$ denote the time t , maturity N forward premium for the

⁶The reference rates are in the Appendix.

⁷Arora, Gandhi, and Longstaff (2012) study the effect of counterparty credit risk on the pricing of one type of swaps, credit default swaps. They show that differences in the credit risk of dealers selling credit protection have a very small effect on the pricing of CDS contracts; in particular, an increase in the credit spread of a dealer of about 645 basis points maps into only a one-basis-point decline in the price of credit protection (the value of the credit protection being sold is diminished). Their sample, however is based on a set of fourteen well known dealers in the U.S. and U.K. who may not be as active in the OIS markets of emerging market economies. Furthermore, identification is based on variation among private dealers; the paper does not quantify the fixed effect arising from the credit risk of private dealers as a whole.

⁸Measuring the price of credit risk of the refreshed interbank panel, and thus, the “risk-free approximation” error, however, requires the elusive risk-free benchmark.

foreign currency vis-a-vis the domestic currency; a positive premium implies a risk-neutral expected depreciation of the domestic currency. Covered interest rate parity implies that the time t , maturity N , nominal foreign OIS rate, which I denote by $y_t^{*,CIP,(N)}$ is

$$y_t^{*,CIP,(N)} = y_t^{(N)} - \rho. \quad (3.36)$$

Du, Tepper, Verdelhan document CIP deviations in interbank lending markets that cannot be explained away by credit risk or transaction costs for G10 currencies. They argue that wide deviations on contracts that are on banks' balance sheets at quarter end suggest regulatory capital costs are driving the wedge at quarter end. This would affect developing market currencies as well. The authors estimate that from 2010 to 2016, the average, annualized basis for currencies in their sample ranged from -60 to 30 bps at the five-year horizon. For context, the level of LIBOR ranged from around 25 bps to 400 bps. They also document heterogeneity across currencies and time, which suggests we would not be able to correct our synthetic South African rand, and Brazilian real OIS rates for these deviations by simple subtraction of a level factor from these rates.

Zero coupon real bond rates To back out zero-coupon real bond rates from market data, I privilege the use of inflation swap prices, over traded real bond prices. The reason is two fold. First, while inflation indexed bonds have existed before 2009, limited issuance across the maturity structure makes it difficult to bootstrap an entire curve of zero coupon rates. In contrast, inflation swap markets have quotes at multiple and evenly spaced maturities. Second, many inflation indexed bonds have embedded options that we would have to price and separate out. For example, U.S. real bonds, formally referred to as U.S. Treasury Inflation-Indexed Securities and informally as U.S. Treasury Inflation Protected Securities (TIPS), have deflation floors to protect the principal value in inflation, whereas comparable U.K. real bonds, formally referred to as U.K. Index-Linked Gilts, do not. This is in contrast to inflation swaps for which the floating leg does not have embedded options.

For all currencies, except for the Brazilian real, I used zero-coupon inflation swaps to compute the zero-coupon real bond rate. In the case of the Brazilian real, I used zero rates published on Bloomberg for Brazilian real bonds; because these bonds do not contain options the zeroes can be obtained from standard bootstrapping. In currencies in which both inflation swaps and real bonds are traded, I privileged the use of inflation swaps rates to obtain zero coupon inflation compensation rates because of their standardization across currencies.

One may wonder whether inflation swap markets are sufficiently liquid, justifying the use of prices in that market as empirical measures of an object referenced in the no arbitrage, pricing relationship described earlier. The literature on the liquidity of inflation swap markets tends to focus on those for the United States and the United Kingdom, in which they have found to be reasonably liquid and the prices reliable. For example, Fleming and Sporn (2019) estimate realized bid-ask spreads for customers are on the order of three basis points. As in Fleming, Longstaff, and Lustig (2014), however, to guard against using stale prices in the sample, I only include an inflation swap rate when the rate has changed from the previous day; that I use data aggregated to the monthly frequency in the empirical work should also mitigate concerns of stale prices.

In constructing zero-coupon real rates from nominal, and inflation compensation rates, I implicitly assume that the inflation swap of a given maturity compensates the floating leg investor with the actual, realized inflation of the swap's lifetime. This is not strictly true because of publication lags in the national price index, and the contractual lag built into inflation swap contracts.

One may wonder how publication lags in the national price index affect the empirical strategy. There are publication lags in the national price index because national price level data take time to collect and construct. The publication lag implies that without perfect forecasting abilities, investors cannot know the true real exchange rate today because the national price levels relevant to its computation have yet to be released. However, because publication lags are less than a month, and real exchange rate changes are computed over a month these concerns should be mitigated.

One may wonder how contractual lags in the swaps affect the empirical strategy. Contractual lags exist because the swap contracts specify cash flows to be exchanged based on lagged values of the inflation index. These lags varies across currencies, but with significantly less variation than for real bonds. For example, U.S. swaps (real bonds) are linked to their inflation index series with a three-month (three-month) lag, whereas U.K. swaps are linked to their inflation index series with a two-month (three to eight-month lag). Their existence implies that an inflation swap is compensation for two types of inflation: known, lagged inflation, and unknown, future inflation. Only the latter will carry an inflation premium. It also implies that at time t , the swap rate on an inflation swap contract for year n and with contractual lag l will be compensation for inflation from $t - l$ to $t + n - l$, instead of what it should be, which is from t to $t + n$. However, since contractual lags are small relative to the 10-year maturity being considered, and existing empirical work on the impact of these lags

on yields, contractual lags likely have minimal effects on inflation swap rates.⁹

National price level series Countries release multiple national price level series; in this study, I use the national price level series referenced by the relevant inflation swap or real bond contract.

Across currencies, the relevant national price level series differ in the real baskets they reference and the timeliness of their releases. For example, the Japanese index references a core basket of real goods that is meant to capture the average consumer's real consumption basket excluding perishables, a sizable but relatively volatile component of consumer consumption baskets, whereas the Swedish index references a headline basket of real goods that includes perishables. Meanwhile, the Brazilian general index of market prices (IGP-M) is not simply the consumer price index, but the arithmetic weighted average of the wholesale price index, consumer price index, and national construction cost index. All series are monthly with the exception of Australia's, which is quarterly and is converted into a monthly series by making a given month's observation the last quarterly observation observed.

⁹Perhaps because of how persistent inflation series are. For example, Christensen (2018), who exploits variation in the indexation lags of U.K. real bonds and estimates the yield difference between two real bonds identical except in that one has a three-month indexation lag, and the other an eight-month indexation lag, is around two to three basis points for ten-year bonds. While the study does not directly address the effect of contractual lags on the measured versus true log real bond holding period returns, if the pricing effect of the inflation lag is around the same order of magnitude, it should not have a large effect on the empirical results. I do not explicitly account for publication or contractual lags in the study.

Table 3.3: National price level index series

Currency	National price index series	Freq.	Publication lag (mo.)	Contract	Contract lag (mo.)
British pound	Retail Price Index	M	0.5	Swap	2
Japanese yen	Consumer Price Index ex perishables	M	1	Swap	3
euro	Harmonized Index of Consumer Prices, All items	M	0.5	Swap	3
Swedish krona	Consumer Price Index	M	0.5	Swap	3
U.S. dollar	Consumer Price Index, All urban consumers	M	0.5	Swap	3
Australian dollar	Consumer Price Index, All groups	Q	1	Swap	3
S. African rand	Consumer Price Index, Metropolitan	M	0.5	Swap	4
Brazilian real	FGV Brazil General Prices IGP-M	M	0.5	Bond	1

Notes: This table reports the national price level series referenced by the zero coupon inflation swap contracts, or in the case of the Brazilian real, the real bond contract. Additionally, it gives the release frequency, the publication lag of the national price level series, and the contractual lag specified in the relevant securities contract. Information was obtained from Bloomberg and national statistical offices.

Nominal versus real log holding period returns

To contextualize this paper in relation to [Lustig et al. \(2019\)](#), which runs the nominal version of the real regression I estimate in this paper, I derive one set of testable assumptions that imply that the nominal and real regressions should be equal.

Proposition 2. *The nominal and real regressions will be equal when Assumptions I-IV hold in nominal and real terms and when the difference in inflation equal the difference in the holding period returns of nominal, long-run inflation swaps:*

$$\pi_{t+1} - \pi_{t+1}^* = hpr_{t+1}^{(10),\pi} - hpr_{t+1}^{*(10),\pi}. \quad (3.37)$$

A special case in which this holds is when log inflation is equal to the log holding period return on the ten-year inflation swap

$$\pi_{t+1} = hpr_{t+1}^{(10),\pi} \quad (3.38)$$

$$\pi_{t+1}^* = hpr_{t+1}^{*,(10),\pi}. \quad (3.39)$$

Proof. If Assumptions I-IV hold in nominal terms, the derivation of the real regression [3.15](#) leads to the derivation of the following nominal regression when all real variables are swapped for nominal variables:

$$\Delta s_{t+1} = hpr_{t+1}^{*,nom,(10)} - hpr_{t+1}^{nom,(10)} + \epsilon_{t+1}, \quad (3.40)$$

where $hpr_{t+1}^{(10),nom}$ is the log holding period return on the long run nominal bond, $hpr_{t+1}^{(10),\pi}$ is the log holding period return on the long run inflation swap and the corresponding variables with stars are the foreign counterparts of the domestic variables just defined.

Recall that

$$\Delta q_{t+1} = hpr_{t+1}^{*,(10)} - hpr_{t+1}^{(10)} + \epsilon_{t+1} \quad (3.41)$$

$$\Delta s_{t+1} + (\pi_{t+1} - \pi_{t+1}^*) = hpr_{t+1}^{*,(10)} - hpr_{t+1}^{(10)} + \epsilon_{t+1} \quad (3.42)$$

where the second line flows from the definition of q_{t+1} . By definition, the log holding period return on the long-run real bond $hpr_{t+1}^{(10)}$ is related to the log holding period return on the long-run nominal bond $hpr_{t+1}^{(10),\pi}$ and the log holding period return on the long-run inflation swap $hpr_{t+1}^{(10),\pi}$:

$$hpr_{t+1}^{(10)} \equiv \log\left(\frac{\left[\frac{(1+i_t^{(9\frac{11}{12})})}{(1+y_t^{(9\frac{11}{12})})}\right]^{9\frac{11}{12}}}{\left[\frac{(1+i_t^{(10)})}{(1+y_t^{(10)})}\right]^{10}}\right) \quad (3.43)$$

$$= \log\left(\frac{\left[\frac{1}{(1+y_t^{(9\frac{11}{12})})}\right]^{9\frac{11}{12}}}{\left[\frac{1}{(1+y_t^{(10)})}\right]^{10}}\right) - \log\left(\frac{\left[\frac{1}{(1+i_t^{(9\frac{11}{12})})}\right]^{9\frac{11}{12}}}{\left[\frac{1}{(1+i_t^{(10)})}\right]^{10}}\right) \quad (3.44)$$

$$= hpr_{t+1}^{(10),nom} - hpr_{t+1}^{(10),\pi}. \quad (3.45)$$

This relationship also holds for the foreign currency. Thus, Eq [3.42](#) can be rewritten as

$$\Delta s_{t+1} + (\pi_{t+1} - \pi_{t+1}^*) = hpr_{t+1}^{*,(10)} - hpr_{t+1}^{(10)} + \epsilon_{t+1} \quad (3.46)$$

$$\Delta s_{t+1} + (\pi_{t+1} - \pi_{t+1}^*) = hpr_{t+1}^{*,(10),nom} - hpr_{t+1}^{*,(10),\pi} - hpr_{t+1}^{(10),nom} + hpr_{t+1}^{(10),\pi} + \epsilon_{t+1} \quad (3.47)$$

$$\Delta s_{t+1} + (\pi_{t+1} - \pi_{t+1}^*) = (hpr_{t+1}^{*,(10),nom} - hpr_{t+1}^{(10),nom}) + (hpr_{t+1}^{(10),\pi} - hpr_{t+1}^{*,(10),\pi}) + \epsilon_{t+1}. \quad (3.48)$$

When the difference in inflation equal the difference in the holding period returns of nominal, long-run inflation swaps:

$$\pi_{t+1} - \pi_{t+1}^* = hpr_{t+1}^{(10),\pi} - hpr_{t+1}^{*,(10),\pi} \quad (3.49)$$

we can further reduce the equation to

$$\Delta s_{t+1} = (hpr_{t+1}^{*,(10),nom} - hpr_{t+1}^{(10),nom}) + \epsilon_{t+1}. \quad (3.50)$$

□

In the data, inflation differentials and ten-year inflation swap holding period return differentials are weakly correlated. In Table [3.3.2](#), I report the regression results of regressing the former on the latter; regression coefficients are near zero, and R^2 are near zero. Thus, the empirics reject a simple testable assumption that would imply that the results of the nominal regression would be identical to the results of the real regression and thus, motivate this paper's empirical exercise.

Table 3.4: Inflation differentials and ten-year inflation swap holding period return differentials

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$(hpr_{t+1}^{(10),\pi} - hpr_{t+1}^{*(10),\pi})$	$(\pi_{t+1} - \pi_{t+1}^*)$	$(\pi_{t+1} - \pi_{t+1}^*)$	$(\pi_{t+1} - \pi_{t+1}^*)$	$(\pi_{t+1} - \pi_{t+1}^*)$	$(\pi_{t+1} - \pi_{t+1}^*)$	$(\pi_{t+1} - \pi_{t+1}^*)$	$(\pi_{t+1} - \pi_{t+1}^*)$
	0.00 (0.02)	-0.06** (0.03)	-0.02 (0.02)	-0.02 (0.02)	-0.02 (0.03)	-0.01 (0.01)	-0.02 (0.03)
Constant	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	-0.00*** (0.00)	-0.00*** (0.00)
R^2	143 0.00	143 0.04	143 0.00	143 0.01	143 0.00	143 0.01	101 0.00
	Japanese yen	euro	Swedish krona	U.S. dollar	Australian dollar	South African rand	Brazilian real
Maturity	10y	10y	10y	10y	10y	10y	10y
Freq.	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly

Notes: Panel A reports the results of a regression of inflation differentials on ten-year inflation swap holding period return differentials.

3.4 Empirical Strategy and Results

3.4.1 Real

Let t index time. Earlier, I showed that Assumption I-IV imply coefficient estimates for the following regression

$$\Delta q_{t+1} = \alpha + \beta_1 hpr_{t+1}^{*,(10)} + \beta_2 hpr_{t+1}^{(10)} + \epsilon_t \quad (3.51)$$

where Δq_{t+1} is the log change in the real exchange rate defined as units of foreign real goods baskets per unit of the domestic real goods basket, $hpr_{t+1}^{*,(10)}$ is the log holding period return on the foreign, ten-year real bond, and $hpr_{t+1}^{(10)}$ is the log holding period return on the domestic, ten-year real bond. If all assumptions, hold, the constant term should equal zero, and the coefficients on the foreign and domestic holding period returns should equal 1 and -1, respectively. This motivates the following hypothesis tests:

$$\text{Test I: } H_0 : \alpha = 0; \beta_1 = 1; \beta_2 = -1 \quad (3.52)$$

$$H_a : \alpha \neq 0; \beta_1 \neq 1; \beta_2 \neq -1. \quad (3.53)$$

I denote this test the “Coefficient Test” and report a rejection if the parameters are jointly insignificant at the 5% level. The Coefficient Test suggests that if the log holding period return on the foreign long-run, real bond increases by 1%, the foreign currency should depreciate contemporaneously by 1%; also, if the holding period return on the domestic, long-run, real bond increases by 1%, the domestic currency should appreciate contemporaneously by 1% appreciation. Thus, using long-run bonds, carry trade returns should be close to zero.

The second testable implication is that the R^2 on the regression is equal to one.

$$\text{Test II: } H_0 : R^2 = 1 \quad (3.54)$$

$$H_a : R^2 < 1. \quad (3.55)$$

I denote this test the “ R^2 Test” and report a rejection if the parameter under the null is not in the confidence set at the 5% level and an acceptance otherwise. Since the parameter under the null is on the boundary of the parameter space it has a non-standard distribution and traditional bootstrap will generate inconsistent estimators. To conduct inference on estimated parameters, I generate confidence intervals by drawing pseudo time series by subsampling (Politis and Romano (1994)). The Appendix contains the algorithm description. The value of this test and of generating confidence intervals based on subsampling in this

setting is questionable because of the small sample size; however, it is the only method that has a chance of being asymptotically consistent. ¹⁰

The regression is valid at all frequencies. Since test power increases with sample size, I maximize the number of observations by running the regression at the monthly frequency, which corresponds to the most common release frequency of the national price level series used to construct real exchange rates.

I report the regression point estimates and test outcomes in Table [3.4.1](#). I describe the results and then discuss their theoretical implications.

Columns 1-5 report results from estimating the regression on the five developed market currencies vis-a-vis the British pound: Japanese yen, euro, Swedish krona, U.S. dollar, Australian dollar. For this set of currencies, the point estimates are of the correct sign: the coefficient on the foreign log holding period return is positive, and the coefficient on the domestic log holding period return is negative. This implies that when there is positive shock to foreign log holding period returns in local currency terms, the foreign currency will tend to contemporaneously depreciate; when there is positive shock to domestic log holding period returns, the foreign currency will tend to contemporaneously appreciate. However, while the coefficient on the domestic return is significant across developed market currencies at the 1% level, the coefficient on the foreign return is only significant at the 1% level for the euro, U.S. dollar, and the Australian dollar.

Columns 6-7 report results from estimating the regression on the two market currencies vis-a-vis the British pound: South African rand and the Brazilian real; Columns 8-9 repeat the regression using holding period returns on the foreign, real bonds that are implied by CIP. For this set of currencies, the point estimates are jointly not of the correct sign when using OIS-based risk-free real bond rates, but are of the correct sign when using CIP-implied risk-free real bond rates. When using OIS-based risk-free real bond rates, the point estimates

¹⁰The value in this context is unclear for two reasons.

The first reason is mechanical: the sample is short, and the block length, which was set to twelve to balance the need for a reasonably sized set of subsamples with the need to keep the subsample length large enough to capture data dependence across time; proofs of subsample estimator consistency, however, rely on double asymptotics: both the sample and block size tend to infinity. As with all proofs that rely on asymptotics, the application of the proof in a setting in which all estimators are derived from finite samples is an act of faith, which may be misplaced.

The second reason is a repeat of Bayesian criticism of frequentist inference that seems particularly relevant in this case. The frequentist 95% confidence interval contains the true parameter with probability 95% before one has seen the data. I have seen the data, and the probability that it is in the confidence interval is exactly zero. The theory is about point-by-point realizations. If the data are all precisely, and correctly measured, a finite sample R^2 less than one is automatic rejection of the hypothesis of point-by-point realizations. Allowing for the possibility that a finite sample R^2 of less than one does not imply automatic rejection requires some idea of why the former might hold, such as measurement error. Suppose the countable, data generating process is such that the null holds true except occasionally. Since the data generating process is countable, in any finite sample, as I take sample size to infinity, the R^2 will approach, but never equal one.

on the domestic log holding period returns are of the correct sign; however, foreign log holding period returns are of the wrong sign: they are negative. This implies that when there is positive shock to foreign log holding period returns, the foreign currency will tend to contemporaneously appreciate. It is possible the sign of the true coefficients are the same as the sign of the estimated coefficients; alternatively, it may be that OIS swaps in developing market economies have sizable and variable risk premia compared to developed market economies that are generating the result.

Adversely towards the null, the coefficients are not one and vary in magnitude across currencies; furthermore, the R^2 on the regressions are far from 1. In particular, the coefficient on foreign returns range from 0.16 to 0.76; the coefficient on domestic returns range from -0.49 to -1.00. The R^2 on the regressions range from 8% for the Swedish krona to 33% for the Brazilian real, with standard errors 18% and 14%, respectively. Thus, at the 5% level, I reject the null hypotheses that the coefficients on the foreign and domestic holding period returns equal to 1 and -1, respectively, as well as the null hypothesis that the R^2 is one. If the domestic OIS rate is truly risk-free, and covered interest parity held, then $hpr_{t+1}^{*,CIP,(10)}$, not $hpr_{t+1}^{*,(10)}$ is the correct empirical counterpart to the ten-year real bond holding period returns in the derivation. Notably, the sign of the coefficient on the foreign holding period bond return reverses signs, and becomes positive (i.e. in the direction of long real bond return parity) and significant, changing from -0.20 to 0.37 for the South African rand, and -0.76 to 0.28 for the Brazilian real. They are, however, still far from the null of 1. The coefficient on the domestic holding period bond return becomes more negative and closer to the null of -1, changing from -0.36 to -0.69 for the South African rand, and from -0.75 to -1.13 for the British real.

Taken together, the results of the real regression estimation uniformly reject the joint hypothesis.

Table 3.5: Benchmark regression using private OIS rates (real)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Δq_{t+1}	Δq_{t+1}	Δq_{t+1}	Δq_{t+1}	Δq_{t+1}	Δq_{t+1}	Δq_{t+1}	Δq_{t+1}	Δq_{t+1}
Panel A: Point estimates									
$hpr_{t+1}^{*,(10)}$	0.16 (0.19)	0.59*** (0.12)	0.21 (0.13)	0.40*** (0.11)	0.61*** (0.14)	-0.20 (0.18)	-0.76*** (0.14)		
$hpr_{t+1}^{*,CIP,(10)}$								0.37*** (0.05)	0.28*** (0.04)
$hpr_{t+1}^{(10)}$	-1.00*** (0.15)	-0.61*** (0.11)	-0.49*** (0.14)	-0.95*** (0.12)	-0.88*** (0.16)	-0.36* (0.21)	-0.75*** (0.23)	-0.69*** (0.18)	-1.13*** (0.22)
Constant	0.00* (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	-0.00 (0.00)	0.00 (0.00)	0.01** (0.00)	0.00 (0.00)	0.01* (0.00)
N. Obs	143	143	143	143	143	143	101	143	101
R^2	0.24	0.19	0.08	0.29	0.19	0.03	0.33	0.27	0.37
R^2 std. dev.	0.22	0.19	0.18	0.20	0.19	0.16	0.14	0.22	0.24
Currency	Japanese yen	euro	Swedish krona	U.S. dollar	Australian dollar	South A. rand	Brazilian real	South A. rand	Brazilian real
Maturity	10y	10y	10y	10y	10y	10y	10y	10y	10y
Freq.	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly
Panel B: Test results									
Coefficient Test	Reject	Reject	Reject	Reject	Reject	Reject	Reject	Reject	Reject
R^2 Test	Reject	Reject	Reject	Reject	Reject	Reject	Reject	Reject	Reject

Notes: Panel A, Columns 1-7 report the results of a regression of the monthly change in the log of real exchange rates denoted by Δq_{t+1} on the contemporaneous holding period return on the real, ten-year foreign bond in foreign currency numeraire $hpr_{t+1}^{*,(10)}$, and the real ten-year domestic bond in domestic currency numeraire $hpr_{t+1}^{(10)}$. Columns 8-9 report the results of a regression of the monthly change in real exchange rates denoted by Δq_{t+1} on the contemporaneous holding period return on the real, CIP implied ten-year foreign bond in foreign currency numeraire $hpr_{t+1}^{*,CIP,(10)}$, and the real ten-year domestic bond in domestic currency numeraire $hpr_{t+1}^{(10)}$ for the South African rand and the Brazilian real. Panel B reports the results of the hypothesis tests. The row Coefficient Test reports on whether I reject or accept the null hypothesis that the coefficient on foreign holding period returns is +1, and the coefficient on domestic holding period returns is -1 at the 5% level. The row R^2 Test reports on whether I reject or accept the null hypothesis that the R^2 is equal to 1 at the 5% level.

3.4.2 Nominal

If the assumptions hold in nominal terms, I can run the regression in nominal terms as well.

Let t index time and i the foreign currency, this directly motivates a regression of the form

$$\Delta s_{t+1} = \alpha + \beta_1 hpr_{t+1}^{*,nom,(10)} + \beta_2 hpr_{t+1}^{nom,(10)} + \epsilon_t \quad (3.56)$$

where Δs_{t+1} is the log change in the nominal exchange rate, $hpr_{t+1}^{*,nom,(10)}$ is the log holding period return on the foreign, ten-year nominal bond, and $hpr_{t+1}^{nom,(10)}$ is the log holding period return on the domestic, ten-year nominal bond. The testable implications are as before.

Table [3.4.2](#) reports regression estimates.

Columns 1-5 reports point estimates from the regression on the five developed market currencies vis-a-vis the British pound: Japanese yen, euro, Swedish krona, U.S. dollar, Australian dollar. For the set of developed market currencies, the point estimates, with the exception of the Japanese yen, are of the correct sign: the coefficient on the foreign log holding period return is positive and the coefficient on the domestic log holding period return is negative.

Comparing these nominal results with the real results, I find that with the exception of the euro and U.S. dollar, the R^2 are higher in the nominal regressions than in the real regressions for the set of currencies. This makes sense because of the weak time series relation between log inflation differentials and log inflation swap holding period return differentials. The two series look as if generated by a process centered around a mean, and subject to weakly correlated mean zero noise innovations. Thus, innovations to inflation differentials will tend to decrease the R^2 in the real regressions vis-a-vis the nominal regressions since it will add variation to the dependent variable not explained by inflation swap holding period return differentials, and this effect appears to dominate for most currencies. The null hypothesis that the coefficients on the foreign and domestic holding period returns equal to 1 and -1, respectively, is accepted for the Swedish krona and the Australian dollar at the 5% level, and rejected for all other currencies. The null hypothesis that the R^2 is one is rejected for all currencies.

Columns 6-7 report results from estimating the regression on the two market currencies vis-a-vis the British pound: South African rand and the Brazilian real;

For the set of developing market currencies, coefficients are of the wrong sign: they are negative for foreign bond returns, and positive for domestic bond returns, implying that a positive shock to foreign (domestic) bond returns are, in domestic currency terms, amplified by foreign currency appreciation (depreciation).

Columns 8-9 re-estimates the regression for developing market currencies using holding period returns on the foreign, real bonds that are implied by CIP. If the domestic OIS rate is truly risk-free, and covered interest parity held, then $hpr_{t+1}^{*,CIP,(10)}$, not $hpr_{t+1}^{*,(10)}$ is the correct empirical counterpart to the ten-year real bond holding period returns in the derivation. Notably, the sign of the coefficient on the foreign holding period bond return reverses signs, and becomes positive (i.e. in the direction of long-run, real bond return parity), changing from -0.69 to 0.58 for the South African rand, and -0.47 to 0.44 for the Brazilian real. They are, however, still far from the null of 1. The coefficient on the domestic bond return also reverses signs, and becomes negative (i.e. in the direction of long real bond return parity) and closer to the null of -1, changing from 0.08 to -0.99 for the South African rand, and from 0.44 to -1.08 for the British real.

Taken together, the results of the real regression estimation uniformly reject the joint hypothesis.

Table 3.6: Benchmark regression using private OIS rates (nominal)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Δs_{t+1}	Δs_{t+1}	Δs_{t+1}	Δs_{t+1}	Δs_{t+1}	Δs_{t+1}	Δs_{t+1}	Δs_{t+1}	Δs_{t+1}
Panel A: Point estimates									
$hpr_{t+1}^{*,nom,(10)}$	-0.96** (0.39)	0.66*** (0.16)	0.91*** (0.17)	0.20 (0.14)	0.87*** (0.14)	-0.69*** (0.10)	-0.47*** (0.06)	0.58*** (0.10)	0.44*** (0.07)
$hpr_{t+1}^{*,nom,CIP,(10)}$								-0.99*** (0.20)	-1.08*** (0.23)
$hpr_{t+1}^{nom,(10)}$	-0.88*** (0.15)	-0.59*** (0.14)	-0.83*** (0.17)	-0.79*** (0.17)	-1.03*** (0.17)	0.08 (0.18)	-0.25 (0.20)	0.01** (0.01)**	0.01*** (0.01)**
Constant	0.01** (0.00)	-0.00 (0.00)	-0.00 (0.00)	0.00 (0.00)	-0.00 (0.00)	0.01** (0.00)	0.01*** (0.00)	0.01** (0.00)	0.01*** (0.00)
N. Obs	143	143	143	143	143	143	101	143	101
R^2	0.33	0.11	0.18	0.21	0.23	0.27	0.38	0.22	0.30
std. dev. of R^2	0.20	0.16	0.17	0.24	0.17	0.22	0.16	0.22	0.20
Currency	Japanese yen	euro	Swedish krona	U.S. dollar	Australian dollar	South African rand	Brazilian real	South African rand	Brazilian real
Maturity	10y	10y	10y	10y	10y	10y	10y	10y	10y
Freq.	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly
Panel B: Test results									
Coefficient Test	Reject	Reject	Accept	Reject	Accept	Reject	Reject	Reject	Reject
R^2 Test	Reject	Reject	Reject	Reject	Reject	Reject	Reject	Reject	Reject

Notes: Panel A Columns 1-7 report the results of a regression of the monthly change in the log of nominal exchange rates denoted by Δs_{t+1} on the contemporaneous log holding period return on the nominal, ten-year foreign bond in foreign currency $hpr_{t+1}^{*,nom,(10)}$ and the nominal, ten-year domestic bond in domestic currency $hpr_{t+1}^{nom,(10)}$. Columns 8-9 report the results of the regression of the monthly log change in nominal exchange rates denoted by Δs_{t+1} on the contemporaneous log holding period return on the nominal, CIP implied ten-year foreign bond in foreign currency $hpr_{t+1}^{*,nom,CIP,(10)}$ and the nominal, ten-year domestic bond in domestic currency $hpr_{t+1}^{nom,(10)}$. Panel B reports the results of the hypothesis tests. The row Coefficient Test reports on whether I reject or accept the null hypothesis that the coefficient on foreign holding period returns is +1, and the coefficient on domestic holding period returns is -1 at the 5% level. The row R^2 Test reports on whether I reject or accept the null hypothesis that the R^2 is equal to 1 at the 5% level.

3.5 Conclusion

In this paper, I considered a specification for real stochastic discount factors in which permanent, martingale components are common across currencies, and tested this assumption jointly with the assumptions that the regularity conditions for the decomposition are satisfied, markets are complete, and ten-year bond returns approximate infinite bond returns. Under these four assumptions, I showed that log real exchange rate changes should be positively correlated with contemporaneous log holding period returns on ten-year foreign real bonds, and negatively correlated with contemporaneous log holding period returns on ten-year domestic bonds; moreover, these relationships should be one-to-one. When using OIS zero rates, the sign of the estimated relationship is correct for developed market currencies, they are not for developing market currencies. Adversely to the joint hypothesis, the magnitude of the relationship, does not align with the theory. A test of the one-to-one relationship and of an R^2 of one is rejected for all currencies by the data. Therefore, I reject the joint hypothesis that all four assumptions jointly hold. An open question is the following: the failure of which assumptions or set of assumptions is primarily responsible for the magnitude of the empirical violation? I leave this for future research.

At the same time, in the post-2009 sample, I document a strong relationship between changes in the log real exchange rate and real holding period bond returns in the direction of parity, contributing to the general question of what explains exchange rate changes (Ferraro, Rossi, and Rogoff (2011), Verdelhan (2018), Jiang, Krishnamurthy, Lustig (2018)).

Bibliography

- ALVAREZ, F. AND U. J. JERMANN (2005): “Using Asset Prices to Measure the Persistence of the Marginal Utility of Wealth,” *Econometrica*, 73, 1977–2016.
- ANDREWS, D. W. K. (2000): “Inconsistency of the Bootstrap when a Parameter is on the Boundary of the Parameter Space,” *Econometrica*, 68, 300–405.
- ARORA, N., P. GANDHI, AND F. A. LONGSTAFF (2012): “Counterparty credit risk and the credit default swap market,” *Journal of Financial Economics*, 103, 280–293.
- AVDJIEV, S., W. DU, C. KOCH, AND H. S. SHIN (2019): “The Dollar, Bank Leverage, and Deviations from Covered Interest Parity,” *American Economic Review: Insights*, 193–208.
- BACKUS, D. K., S. FORESI, AND C. I. TELMER (2001): “Affine Term Structure Models and the Forward Premium,” *Journal of Finance*, 279–304.
- CHRISTENSEN, J. H. (2018): “Do Adjustment Lags Matter for Inflation-Indexed Bonds?” Tech. rep., Federal Reserve Bank of San Francisco.
- DU, W., J. IM, AND J. SCHREGER (2018a): “The U.S. Treasury Premium,” *Journal of International Economics*, 112, 167–181.
- DU, W. AND J. SCHREGER (2016): “Local Currency Sovereign Risk,” *Journal of Finance*, 71, 1027–1070.
- DU, W., A. VERDELHAN, AND A. TEPPER (2018b): “Deviations from Covered Interest Rate Parity,” *Journal of Finance*, 73(3), 915–957.
- FERRARO, D., K. ROGOFF, AND B. ROSSI (2015): “Can Oil Prices Forecast Exchange Rates?” *Journal of International Money and Finance*, 116–141.
- FILIPOVIC, D. AND A. B. TROLLE (2013): “The term structure of interbank risk,” *Journal of Financial Economics*, 109(3), 707–733.
- FLECKENSTEIN, M., F. A. LONGSTAFF, AND H. LUSTIG (2013): “Deflation Risk,” Working paper, NBER.
- FLECKSTEIN, M., F. A. LONGSTAFF, AND H. LUSTIG (2014): “The TIPS-Treasury Bond Puzzle,” *The Journal of Finance*, 69, 2151–2197.
- FLEMING, M. J. AND J. R. SPORN (2013): “Trading Activity and Price Transparency in the Inflation Swap Market,” *FRBNY Economic Policy Review*.

- HULL, J. AND A. WHITE (2015): “OIS Discounting, Interest Rate Derivatives, and the Modeling of Stochastic Interest Rate Spreads,” *Journal of Investment Management*, 13, 64–83.
- JIANG, Z., A. KRISHNAMURTHY, AND H. LUSTIG (2018): “Foreign Safe Asset Demand and the Dollar Exchange Rate,” Tech. rep., NBER.
- JOHANNES, M. AND S. SUNDARESAN (2007): “The Impact of Collateralization on Swap Rates,” *Journal of Finance*, 62(1), 383–410.
- LUSTIG, H., A. STATHOPOULOS, AND A. VERDELHAN (2019): “The Term Structure of Currency Carry Trade Risk Premia,” *American Economic Review*, 109, 4142–77.
- LUSTIG, H. AND A. VERDELHAN (2007): “The Cross Section of Foreign Currency Risk Premia and Consumption Growth Risk,” *American Economic Review*, 97, 89–117.
- MEESE, R. AND K. ROGOFF (1983): “Empirical Exchange Rate Models of the Seventies: Do They Fit Out of Sample,” *Journal of International Economics*.
- POLITIS, D. N. AND J. P. ROMANO (1992): “A Circular Block Resampling Procedure for Stationary Data,” in *Exploring the Limits of Bootstrap*.
- (1994): “Large Sample Confidence Regions Based on Subsamples under Minimal Assumptions,” *Annals of Statistics*, 22, 2031–2050.
- TORTORICE, D. L. AND A. KITA (2018): “Can Risk Models Extract Inflation Expectations from Financial Market Data? Evidence from the Inflation Protected Securities of Six Countries,” Working paper, College of the Holy Cross, Department of Economics.
- VERDELHAN, A. (2018): “The Share of Systematic Variation in Bilateral Exchange Rates,” *Journal of Finance*.

Appendix

3.5.1 Alvarez and Jermann decomposition regularity conditions

The regularity conditions in [Alvarez and Jermann \(2005\)](#) are described below.

First, there exists some δ such that

$$0 < \lim_{k \rightarrow \infty} \frac{P_t^{(k)}}{\delta^k} < \infty, \forall t \quad (3.57)$$

which requires the average yield $-1/kP_t^{(k)}$ converges fast enough to a finite, positive value. Second, for each $t+1$ there exist a random variable with finite expected value $E_t(x_{t+1})$ such that $\forall k$ almost surely

$$\frac{\Lambda_{t+1}}{\delta^{t+1}} \frac{P_{t+1}^{(k)}}{\delta^k} \leq x_{t+1} \quad (3.58)$$

which says the marginal valuation multiplied by $\frac{P_{t+1}^{(k)}}{\delta^k}$ is bounded by a variable that has a finite conditional expectation.

They also maintain the background assumption that stochastic discount factors are strictly stationary. That is, the random vectors

$$\left\{ \frac{\Lambda_{t+1+k}}{\Lambda_{t+k}} \right\}_{k=i}^{k=i+j}, \left\{ \frac{\Lambda_{t+1+k}}{\Lambda_{t+k}} \right\}_{k=l}^{k=l+j} \quad (3.59)$$

should have the same joint distribution $\forall i, j, l$. This implies that all bond returns are stationary.

3.5.2 Drawing pseudo time series

Subsampling

Unlike bootstrap, the resample size is smaller than the sample size and resampling is done without replacement. In the simple i.i.d. case, I generate q equals n choose b blocks of size b from the data, where n is data size I generate the bootstrapped statistic with each block of data. I then use the empirical distribution over the q pseudo estimate, where the empirical estimate is treated as if the true mean, to approximate the distribution of the empirical estimate. With time series, one must respect the ordering of the data to preserve time series dependence, so I draw blocks with contiguous observations.

The results about estimates from subsampling come from proofs that take $b, n \rightarrow \infty, \frac{b}{n} \rightarrow \infty$. In implementation, I am faced with a finite sample, and thus, constrained to a finite b . The asymptotic conditions provide little guidance on how to choose b in this case; as Politis, Romano, and Wolf note, the choice depends on the purpose. I wish in subsampling to preserve time series dependence, and based on autocorrelation charts, choose a block length of fourteen.

Han Hong's primer on subsampling is available here:

<https://web.stanford.edu/~doubleh/lecturenotes/lecture13.pdf>

Monte-Carlo and stationary block bootstrap

The Politis, Romano (1992) algorithm for the stationary block bootstrap is a way for sampling the data in a way that preserves any time dependence in the data, and stationary. It works as follows. Suppose one has a dataset of length N , and wishes to generate nB pseudo time series. First, one splices the initial data into a series of N blocks. In constructing these blocks, the data are treated as circular so that the next observation after the “last” observation is the “first observation.” Two sets of indices of length nB are then randomly generated. The first set of indices is drawn from a uniform distribution. The second set of indices is drawn from a geometric distribution. To generate a pseudo time series, you “choose” and then “stitch together.” The choosing is as follows: the first index determines the block number to choose, the second index determines the length of continuous observations to take from that block number.

Both the Monte Carlo algorithm and the Politis, Romano (1992) algorithm can be implemented by installing the R package `tseries`, and using the function `tsbootstrap`. The first corresponds to the block type bootstrap where the length of the block is equal to one. The second corresponds to the block type stationary where the length of the mean block is equal to what is desired.

Cross section

The motivation for testing the assumption of common permanent components in real currency stochastic discount factors was the finding by Lustig et al. (2019) that average excess returns on the carry trade using long-run bonds and with a three-month horizon, is zero, suggesting that the entropies of the permanent components are equal. Their result is in nominal terms and on currency portfolios. They argue that the inflation characteristics of portfolios suggest the result holds also in real terms. I find suggestive evidence that this is true when I look at the individual currencies in the sample.

3.5.3 Estimated average excess returns by currency

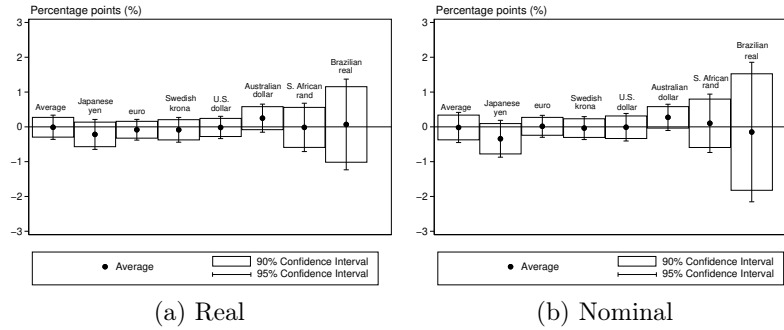
For each currency i , I estimate the average log excess return $\bar{x}r_{t+1}^{i,(10)}$ of going long the ten-year foreign bond and short the ten-year domestic bond expressed in the domestic numeraire (i.e. the British pound basket):

$$\bar{x}r_{t+1}^{i,(10)} \equiv \hat{\mathbb{E}}[hpr_{t+1}^{i,(10)} - \Delta q_{t+1} - hpr_{t+1}^{(10)}]. \quad (3.60)$$

I do this on both a real and nominal basis, and plot the estimated averaged by currency in Figure 3.1. The dots denote the point estimates, the bars denote the 90% confidence interval, and the whiskers denote the 95% confidence interval. Confidence intervals were constructed using standard errors that take into account heteroskedasticity and autocorrelation.

Real, average excess returns range from -22 basis points to 32 basis points, which translates in annualized terms to 2.64 and 3.84 percentage points. Symmetric, 90% and 95% confidence intervals around those point estimates, which are denoted by bars and whiskers respectively, include zero; that is, the estimates are not statistically different from zero. The nominal, average excess returns paint a similar picture: estimates of the average are not statistically different from zero. Confidence intervals on estimates are slightly wider than their real counterparts for the Swedish krona and the Australian dollar.

Figure 3.1: Average excess returns on long-maturity bonds



Note: This plot displays the average real and nominal log monthly excess returns on long maturity (i.e. ten-year) bonds. The domestic currency is the British pound, and the real numeraire good is the British pound basket. The dots denote the average, the bars denote the 90% confidence interval, and the whiskers denote the 95% confidence interval. Confidence intervals were constructed using standard errors that take into account heteroskedasticity and autocorrelation. Returns are monthly, holding period returns (i.e. not annualized), reported in percentage points (i.e. $\log(1.05) \approx 5\%$ is reported as 5, not as .05), and cover February 2009 to December 2020 with the exception of the Brazilian real, whose series begins July 2012.

3.5.4 Decomposition

Before I turn to the time series, I present a decomposition of the average real excess return in Table [3.5.4](#). When I decompose the average excess returns on long-maturity bonds, I see that the change in real exchange rates is offsetting the change in the difference in local currency holding period returns. That is, average excess returns are not zero simply because real exchange rate changes, and local currency holding period returns are zero in expectation.

Table 3.7: Decomposition of the average real excess return on long-maturity bonds

	Currency				
	Average	Japanese yen	euro	Swedish krona	U.S. dollar
(real returns, in percentage points)	0.29	0.24	0.25	0.25	0.22
Foreign holding period bond return (A)	0.20	0.20	0.20	0.20	0.20
Domestic holding period bond return (B)					
Foreign minus domestic holding period bond return (C=A-B)	0.09	0.04	0.05	0.05	0.02
FX return (D)	0.09	0.26	0.13	0.09	0.03
Excess returns in domestic currency (E= C – D)	0.00	-0.22	-0.08	-0.05	-0.02
HAC standard errors (F)	0.18	0.22	0.15	0.18	0.16
No. of monthly return observations	143	143	143	143	143
Sample start	2009-Feb	2009-Feb	2009-Feb	2009-Feb	2009-Feb
Sample end	2020-Dec	2020-Dec	2020-Dec	2020-Dec	2020-Dec

Notes: This table reports the decomposition of the average real excess return on long-maturity bonds. The domestic currency is the British pound, and the real numeraire good is the British pound basket.

Table 3.8: Decomposition of the average real excess return on long-maturity bonds, continued

(real returns, in percentage points)	Currency		
	Australian dollar	South African rand	Brazilian real
Foreign holding period bond return (A)	0.34	0.22	0.56
Domestic holding period bond return (B)	0.20	0.20	0.13
Foreign minus domestic holding period bond return (C=A-B)	0.14	0.02	0.43
FX return (D)	-0.11	0.03	0.36
Excess returns in domestic currency (E= C - D)	0.25	-0.01	0.07
HAC standard errors (F)	0.21	0.36	0.66
No. of monthly return observations	143	143	101
Sample start	2009-Feb	2009-Feb	2012-Jul
Sample end	2020-Dec	2020-Dec	2020-Dec

Notes: This table reports the decomposition of the average real excess return on long-maturity bonds. The domestic currency is the British pound, and the real numeraire good is the British pound basket.

3.5.5 Cross sectional result

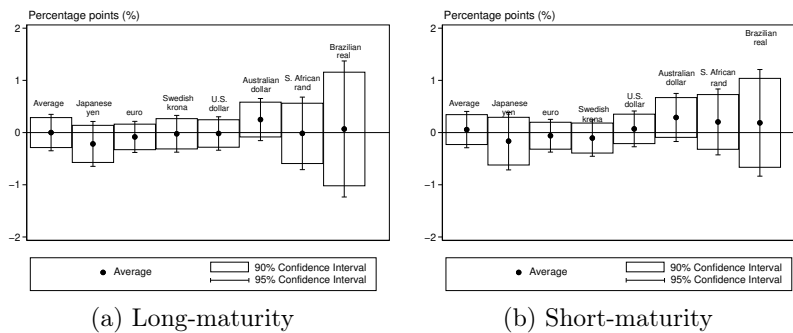
Term structure

To put the long-maturity result in context, I present the corresponding short-maturity result.

Inflation swap data exist only for maturities greater than one year, limiting us to the study of returns on bonds of maturities greater than a year. Therefore, I am unable to estimate the unconditional exchange rate premium as conventionally measured, which is the average monthly bond excess returns on short-run bonds where short-run is the horizon of the trade, in this case one-month. Instead, I estimate the average monthly bond excess returns on short-maturity bonds, defined as two-year bonds, which I plot in Figure 3.5.5. I find that excess returns on short-maturity range from -39 basis points, and 36 basis points, but as before, they are not statistically different from zero.

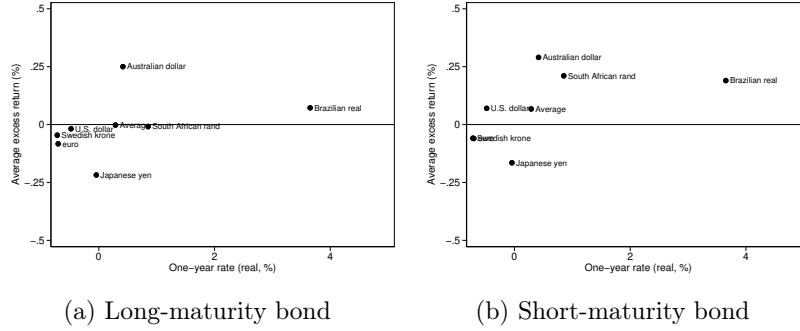
The inability to statistically distinguish from zero is due both to the small magnitude of the point estimates, and large standard errors that reflect the short length of the time series. The point estimates of average excess returns, however, do display a linear relationship vis-a-vis the level of interest rates and the yield curve slopes. Figure 3.5.5 plots the average excess return on the long-maturity and short-maturity bonds vis-a-vis the level of real interest rates. This relationship gets slightly stronger for short-maturity bonds, principally because of the observations for the South African rand and Brazilian real.

Figure 3.2: Average real excess returns on long and short-maturity bonds



Note: This plot displays the average real log monthly excess returns on long maturity (i.e. ten-year) bonds versus short maturity (i.e. two-year) bonds. The domestic currency is the British pound, and the real numeraire good is the British pound basket. Returns are monthly, log holding period returns, reported in percentage points, and cover February 2009 to December 2020 with the exception of the Brazilian real, whose series begins July 2012. Returns for the currency Average was constructed by taking the equally weighted return of currencies with return observations.

Figure 3.3: Average excess returns with the long and short-maturity bond versus the one-year rate



Note: This plot displays the average real log monthly excess returns on long maturity (i.e. ten-year) bonds versus the one-year rate, and the the yield curve slope of the foreign currency. The one-year rate is the one-year, real rate in local currency. The slope is the ten-year minus the one-year real rate in local currency. The domestic currency is the British pound, and the real numeraire good is the British pound basket. Returns are monthly, log holding period returns, reported in percentage points, and cover February 2009 to December 2020 with the exception of the Brazilian real, whose series begins July 2012. Returns for the currency Average was constructed by taking the equally weighted return of currencies with return observations.

3.5.6 Reference interbank rates on plain vanilla OIS contracts by currency

The reference interbank rate on the OIS is typically the average of interest rates at which banks offer to lend unsecured funds to other banks in the interbank market, and is often but not always, an indicative and not market rate; for the Australian dollar the corresponding the rate is the interbank overnight cash rate, and is calculated as the weighted average of the interest rate at which overnight unsecured funds are transacted in the domestic interbank market; for the Brazilian real is the Brazil Interbank Deposit Rate Annualized also known as the CDI, which is the average overnight dollar deposit rate published by the central OTC and private securities and derivatives despoitory CETIP; for the euro is the euro overnight interbank interest rate also known as EONIA; for the British pound is the sterling overnight index average also known as SONIA; for the Japanese yen is the overnight call rate, for the Swedish krona is the tonight next (T/N) Swedish interbank offered rate also known as T/N Stibor; for the U.S. dollar is the interbank interest rate for satisfaction of reserve requirements also known as the fed funds rates.

3.5.7 Other supplementary charts and tables

Table 3.9: Inflation and ten-year inflation swap holding period returns

	(1)	(2)	(3)	(4)	(5)
$hpr_{t+1}^{(10),\pi}$	π_{t+1}	π_{t+1}^*	π_{t+1}^*	π_{t+1}^*	π_{t+1}^*
	0.00				
	(0.02)				
$hpr_{t+1}^{*(10),\pi}$		-0.05***	-0.07	-0.03	-0.02
		(0.02)	(0.05)	(0.02)	(0.02)
Constant	0.00***	0.00	0.00***	0.00***	0.00***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
	143	143	143	143	143
R^2	0.00	0.06	0.02	0.01	0.01
0.01	British pound	Japanese yen	euro	Swedish krona	U.S. dollar
Maturity	10y	10y	10y	10y	10y
Freq.	Monthly	Monthly	Monthly	Monthly	Monthly

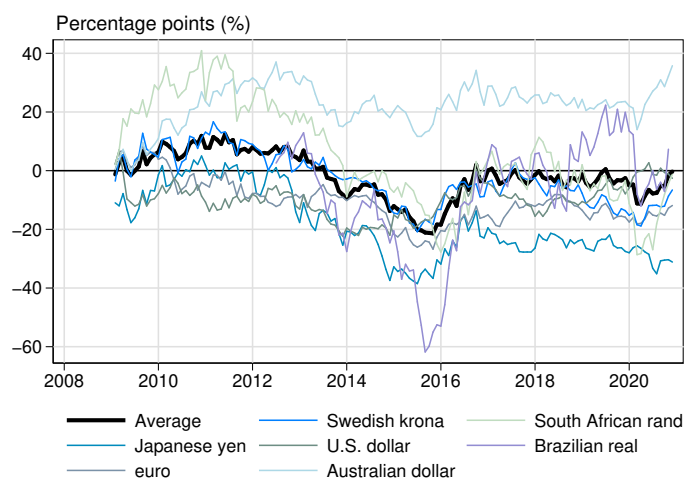
Notes: Panel A reports the results of a regression of inflation on ten-year inflation swap holding period returns.

Table 3.10: Inflation and ten-year inflation swap holding period returns, continued

	(6)	(7)	(8)
$hpr_{t+1}^{(10),\pi}$	π_{t+1}^*	π_{t+1}^*	π_{t+1}^*
$hpr_{t+1}^{*(10),\pi}$	0.03 (0.03)	-0.02 (0.01)	-0.03 (0.03)
Constant	0.00*** (0.00)	0.00*** (0.00)	0.01*** (0.00)
R^2	143 0.01	143 0.01	101 0.01
0.01	Australian dollar	South African rand	Brazilian real
Maturity	10y	10y	10y
Freq.	Monthly	Monthly	Monthly

Notes: Panel A reports the results of a regression of inflation on ten-year inflation swap holding period returns.

Figure 3.4: Cumulative real excess returns on the foreign long-maturity bond



Notes: This plot displays the cumulative, log monthly real excess returns of going long the long-maturity (i.e. ten-year) foreign bond and short the domestic bond. The domestic currency is the British pound, and the real numeraire good is the British pound basket. Standard errors take into account heteroskedasticity and autocorrelation. Returns are monthly, log holding period returns, reported in percentage points, and cover February 2009 to December 2020 with the exception of the Brazilian real, whose series begins July 2012. Returns for the currency Average was constructed by taking the equally weighted return of currencies with return observations.